# 2D Blackboard-Style Case: Start/Target at Wall Endpoints

## Summary

- **Start/target at the corridor endpoints**: $S = (30, 45)$ and $T = (30, 15)$, with the same two-wall corridor geometry.
- **No bimodality**: $t_{p1} \approx 102$, valley ratio $= 1$ (single peak + long tail); absorption probability $\approx 0.604$ at $t_{\max} = 3000$.
- **Small $N$ vertical rectangle also fails**: on an $18 \times 30$ rectangle with the same endpoint placement, $t_{p1} \approx 40$, valley ratio $= 1$, absorption $\approx 0.664$ at $t_{\max} = 2000$.
- **Scan conclusion**: changing $g_y$, the opening-arrow strength, or adding only an outer sticky belt does not create a second peak; a late peak appears only after adding an *outer sticky + directional loop* (an added slow channel).
- **Full figure set included**: environment, three heatmap frames, fast/slow path density, FPT curve + zoom, diagnostic plot, and channel decomposition.

## Global parameters

| case | $N$ | $q$ | $g_x$ | $g_y$ | $t_{\max}$ | (start, target) |
|---|---|---|---|---|---|---|
| Z | 60 | 0.8 | 0.00 | -0.20 | 3000 | (30,45)→(30,15) |
| S | 30 | 0.8 | 0.00 | -0.20 | 2000 | (9,22)→(9,9) |

Table 1: Global parameters for the wall-endpoint case.

## Statistics

| case | method | $t_{p1}$ | $t_{p2}$ | valley ratio | absorption | truncated (MC) | $P_{\text{fast}}$ (MC) |
|---|---|---|---|---|---|---|---|
| Z | windowed | 102 | 1500 | 1.000 | 0.604 | 0.395 | 0.540 |
| S | windowed | 40 | 1000 | 1.000 | 0.664 | 0.333 | 0.543 |

Table 2: Statistics for the wall-endpoint case. $t_{p2}$ is the late-window maximum (not a genuine second peak).

## Numerical scans (based on case Z)

**Scan takeaways** Z0–Z4 remain single-peak + long tail; the late-window "peak" is a window-edge artifact. Adding only a sticky belt does not induce bimodality; a clear second peak appears only when an outer directional loop is introduced (Z5/Z6), which effectively creates a new slow channel.

| variant | modification | absorption | $t_{p1}$ | $t_{p2}$ | valley ratio | bimodal |
|---|---|---|---|---|---|---|
| Z0 | base ($g_y = -0.2,\ \delta_{\text{open}} = 0.6$) | 0.604 | 102 | 1500 | 1.000 | no |
| Z1 | $g_y = 0$ | 0.696 | 137 | 1500 | 1.000 | no |
| Z2 | $g_y = -0.4$ | 0.573 | 81 | 1500 | 1.000 | no |
| Z3 | $\delta_{\text{open}} = 0.2$ | 0.611 | 102 | 1500 | 1.000 | no |
| Z4 | outer sticky belt ($\alpha = 0.2$) | 0.603 | 102 | 1500 | 1.000 | no |
| Z5 | sticky ($\alpha = 0.2$) + clockwise loop bias ($\delta = 0.1$) | 0.603 | 102 | 2996 | 0.300 | yes |
| Z6 | stronger sticky ($\alpha = 0.05$) + clockwise loop bias ($\delta = 0.1$) | 0.623 | 102 | 1990 | 0.014 | yes |

Table 3: Scan results at $t_{\max} = 3000$. For Z0–Z4, $t_{p2}$ is the late-window maximum (not a genuine second peak).

# 1 Model definition and coordinates

The state space is $\Omega \subseteq \{1, \ldots, N_x\} \times \{1, \ldots, N_y\}$ (rectangles are allowed), and the target $(x_t, y_t)$ is absorbing. One-step probabilities are

$$p_{\text{left}} = \frac{q}{4}(1 + g_x), \quad p_{\text{right}} = \frac{q}{4}(1 - g_x), \tag{1}$$

$$p_{\text{down}} = \frac{q}{4}(1 + g_y), \quad p_{\text{up}} = \frac{q}{4}(1 - g_y), \tag{2}$$

$$p_{\text{stay}} = 1 - q. \tag{3}$$

We use the convention "$y$ increases downward," so $g_y > 0$ biases downward and $g_x > 0$ biases leftward. Both the outer boundary and internal walls are **fully reflecting**: any attempted move outside $\Omega$ or across a wall is reassigned to $p_{\text{stay}}$.

**Local heterogeneity (composition rules)**

- local bias (arrow): multiply $p_{\text{stay}}$ by $(1 - \delta)$ and add $\Delta = \delta\, p_{\text{stay}}$ to the arrow direction.
- sticky: set $q_u = \alpha(u)q$ so $p_{\text{stay}}(u) = 1 - q_u$ is larger, then apply local bias.
- barrier: remove edge $u \leftrightarrow v$ and add its probability to $p_{\text{stay}}(u)$.
- door: pass with probability $p_{\text{pass}}$, reflect the remainder to $p_{\text{stay}}(u)$.

# 2 Exact FPT recursion

Define $\tau = \inf\{t \geq 1 :\ X_t = (x_t, y_t)\}$, $f(t) = \Pr(\tau = t)$. Let $Q$ be the transient submatrix and $r$ the one-step flow into the absorbing state, then

$$f(t) = r^\top Q^{t-1} p_0, \qquad t \geq 1,$$

with the equivalent recursion $p_{t+1} = Qp_t$, $f(t) = r^\top p_t$. Exact recursion is the baseline; AW inversion and MC are used for validation and channel statistics.

# 3 Figure legend

Figure 1 shows the unified symbols: red square = start, blue diamond = target; solid thick lines = reflecting walls, dashed lines = doors, translucent shading = sticky region; red arrows = local bias; the outer arrow gives the global bias $(g_x, g_y)$. Channel cartoon is in Fig. 2.

# 4 Case Z: Start/target at wall endpoints

**Geometry and parameters**

- Internal walls: $x = 28 \leftrightarrow 29$ and $x = 31 \leftrightarrow 32$ for $y = 15..45$.
- Corridor bias: upward arrows on $x = 29..31$; core segment $y = 15..45$ uses $\delta_{\text{core}} = 0.95$, open segments $y = 10..14, 46..50$ use $\delta_{\text{open}} = 0.6$.
- Start/target: $S = (30, 45)$, $T = (30, 15)$; $g = (0, -0.2)$, $t_{\max} = 3000$.

Figure 1: Symbol legend.

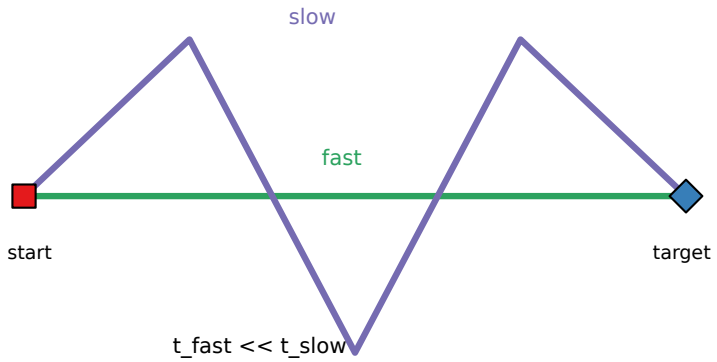
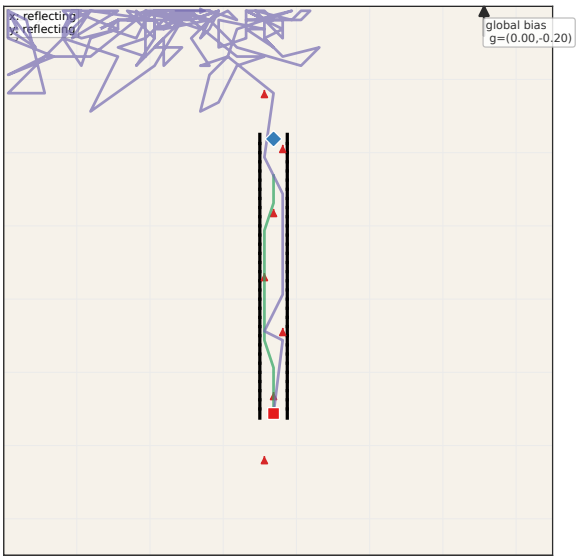


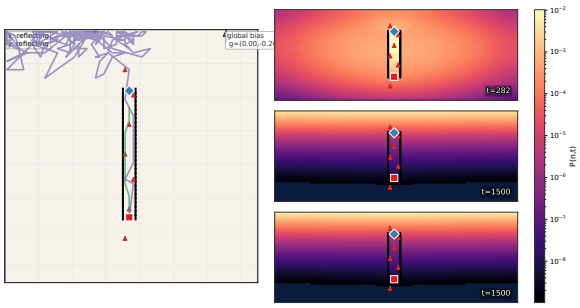

Figure 2: Channel cartoon (fast/slow mixture).

**Outcome** Early peak at $t \approx 102$, valley ratio $= 1$; absorption probability is $\approx 0.604$ at $t_{\max} = 3000$, so no stable late peak appears.

**Why there is no bimodality** With both endpoints inside the corridor, the strong upward corridor bias defines the dominant drift timescale. Trajectories that exit the corridor do not enter a *quasi-closed* slow channel; they diffuse broadly in the outer domain and eventually return, producing a long tail rather than a concentrated late-time peak. In mixture terms, the "slow" conditional distribution is too broad to carve out a valley and a second peak in $f(t)$.

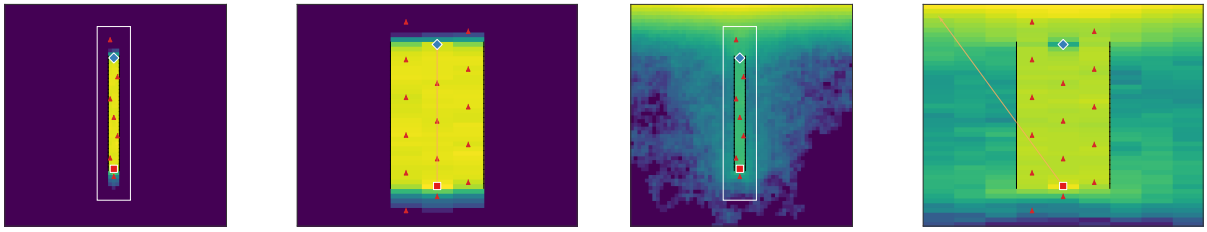



(Z1) Environment



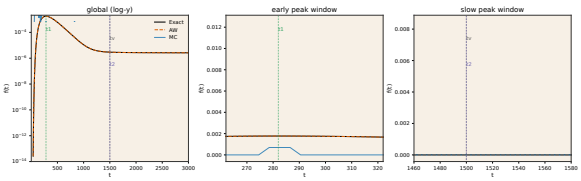

(Z2) Heatmaps

Figure 3: Case Z environment and $P(n, t)$ heatmaps.

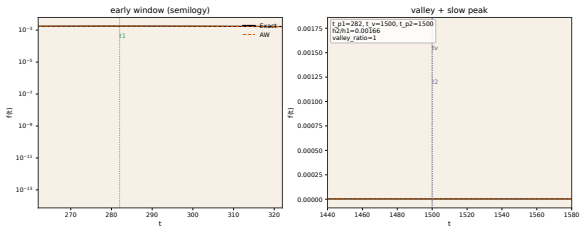


(Z3) Fast-path density (Z4) Slow-path density

Figure 4: Case Z fast/slow path densities.

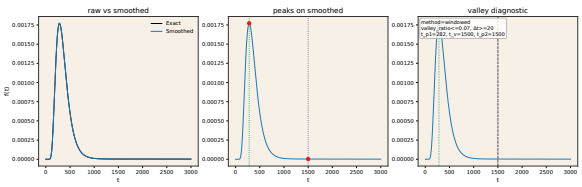



(Z5) FPT curve

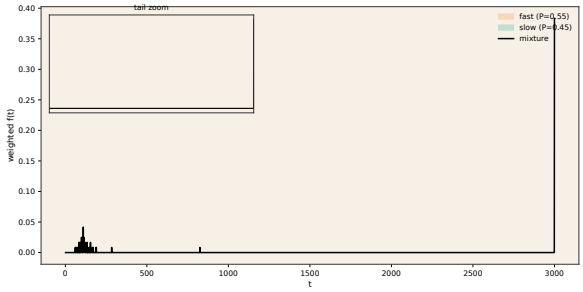



(Z6) Peak/valley zoom

Figure 5: Case Z FPT and zoomed view.




(Z7) Diagnostic




(Z8) Channel decomposition

Figure 6: Case Z diagnostics and channel decomposition.

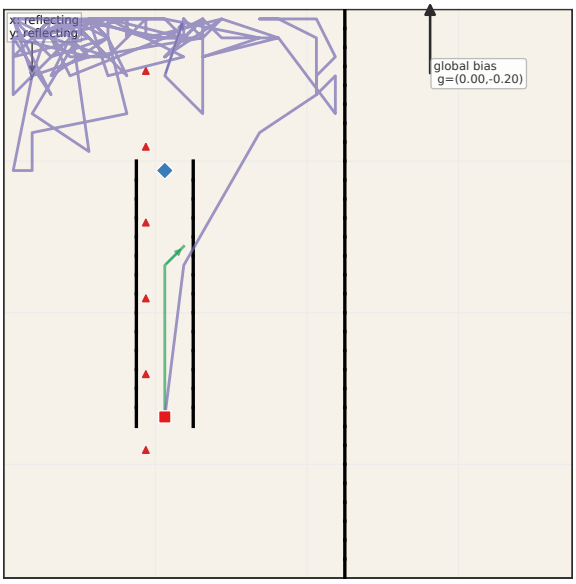
## 5 Case S: Small vertical rectangle (endpoints on walls)

**Geometry and parameters**

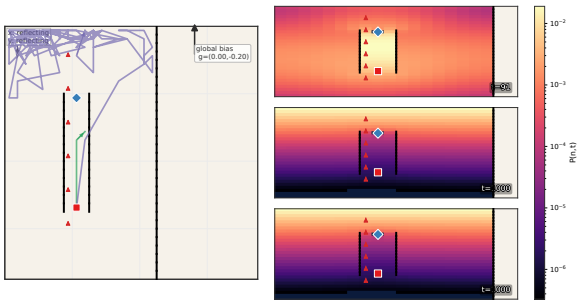
- Domain: $18 \times 30$ rectangle (allowed set $x = 1..18,\ y = 1..30$ embedded in an $N = 30$ grid).
- Internal walls: $x = 7 \leftrightarrow 8$ and $x = 10 \leftrightarrow 11$ for $y = 9..22$.
- Corridor bias: upward arrows on $x = 8..10$; core segment $y = 9..22$ uses $\delta_{\text{core}} = 0.95$, open segments $y = 4..8, 23..27$ use $\delta_{\text{open}} = 0.6$.
- Start/target: $S = (9, 22),\ T = (9, 9)$; $g = (0, -0.2)$, $t_{\max} = 2000$.

**Outcome** Early peak at $t \approx 40$, valley ratio $= 1$; absorption probability is $\approx 0.664$ at $t_{\max} = 2000$, so no stable late peak appears.

**Mechanism** Shrinking to a vertical rectangle does not create a slow channel. Excursions outside the corridor diffuse broadly and return, producing a long tail rather than a concentrated late-time peak.



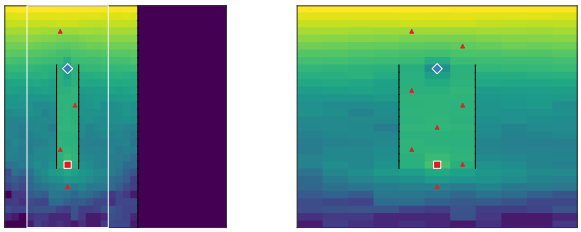

(S1) Environment



(S2) Heatmap frames

Figure 7: Case S environment and $P(n, t)$ heatmaps.

(S3) Fast-path density

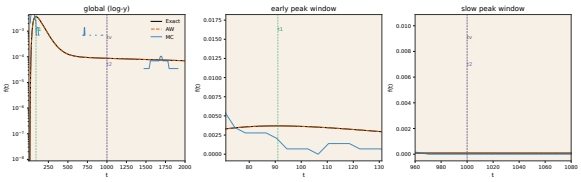


(S4) Slow-path density

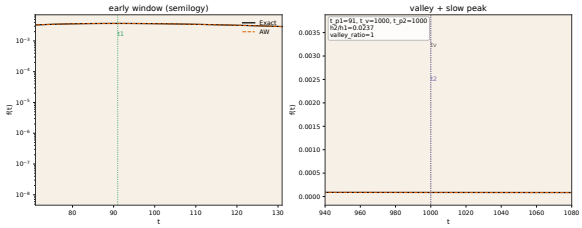
Figure 8: Case S fast/slow path densities.

## Data and reproduction

```
cd reports/grid2d_blackboard_bimodality
python3 code/blackboard_bimodality_pipeline.py --cases Z,S
python3 code/z_scan.py
```

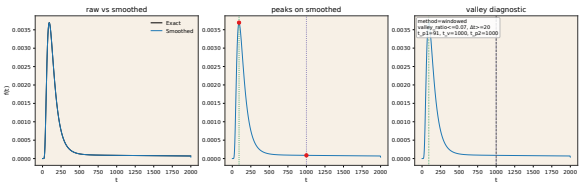



(S5) Multiscale FPT

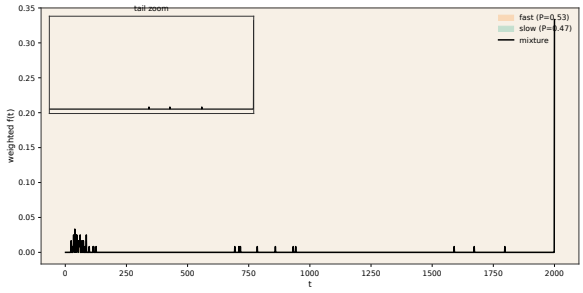



(S6) Peak/valley zoom

Figure 9: Case S FPT curve and zoom.




(S7) Diagnostic




(S8) Channel decomposition

Figure 10: Case S diagnostics and channel decomposition.

```
latexmk -xelatex -auxdir=build -emulate-aux-dir
↪  grid2d_blackboard_bimodality_cn.tex
latexmk -pdf -auxdir=build -emulate-aux-dir
↪  grid2d_blackboard_bimodality_en.tex
```